

Quantum Algorithm to Verify if There Are Infinite Sophie Germain Primes

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Abstract

We give a first quantum algorithm for verifying if there are an infinite number of Sophie Germain primes or if there are a finite number of Sophie Germain primes. This conjecture has been known for over a hundred and fifty years. Our quantum algorithm gives an exponential speed up, compared to a classical version of the algorithm, depending on if there are an infinite number of Sophie Germain primes (at their conjectured density) or a finite number of Sophie Germain primes. Our quantum algorithm, remarkably, requires only one quantum function evaluation instead of an exponential number of classical function evaluations on some interval.

1 Introduction

The SGP (Sophie-Germain-Primes) conjecture is the following (which has been known for over 150 years).

SGP Conjecture. There are an infinite number of Sophie Germain Primes.

If this conjecture is false then there are no Sophie Germain Prime after some (large) integer. It is a well known conjecture in number theory, The infinitude of SGPs is predicted by the Cramer probabilistic model for prime numbers. Since according to the Cramer's model the probability that n is prime is $1/\ln n$, and the probability that $2n + 1$ is prime is $1/\ln(2n + 1)$. Hence the joint probability is

$$(1/\ln(n)) * 1/\ln(n) \approx 1/\ln^2(n)$$

Thus the expected number of SGP predicted by Cramers model is

$$\sum_{n=2}^x \frac{1}{\ln^2(n)} \sim li_2(x) \quad (2)$$

But the integral $li_2(x)$ diverges as $x \rightarrow \infty$ so Cramers model predicts an infinitude of SGPs. The above is equivalent to saying the Cramer's model predicts the gap between consecutive SGP is $O(\ln^2(n))$. Let SGP_n mean the n th SGP prime. It turns out that

$$SGP_{n+1} - SGP_n \leq \lceil 8 \ln(n) \rceil^2 \quad (3)$$

works for $O(\ln^2(n)) = \lceil 8 \ln(n) \rceil^2$; this bound is not tight but it works for every known pair, which are enormous numbers, of consecutive SGP. While the bound 3 above is not tight it is sufficient for using in our new algorithm, and there exists a tighter bound than 3 above we could use instead if we chose to. So it is reasonable to conjecture the bound $\lceil 8 \ln(n) \rceil^2$ to be true as it is supported by Cramer's model. It is also true that if inequality 3 holds infinitely often then it supports the prediction by Cramers model that there are infinite SGPs with gap distance $O(\ln^2(n))$.

Therefore we can test the conjecture that there are an infinite number of SGPs by using the bound 3 above up to some large number. If the conjecture is false, then this means there will be no SGPs after some very large number then 3 will be false. The question of if there are an infinite number of SGP has been an open number in number theory for over 150 years-it a very old problem like factoring integers (which was studied by Gauss). Like factoring integers we give a polynomial time and polynomial space quantum algorithm to solve this problem; it is an exponential speed up for the problem and actually the exponential speed up is greater than the Shor algorithm (though the Shor algorithm is for different problem in number theory). It is interesting to note that testing if there are an infinite number of SGP on some interval is a problem in NP, in this paper we show that testing the conjecture that there are an infinite number of SGP on an exponential sized interval is in BQP (Bound-error Quantum Polynomial). Interestingly, our polynomial time quantum algorithm is not general quantum Fourier based (since it is Deutsch-Jozsa based) but it is interference based. Factoring integers classically is suspected to be a problem in NP and not in P. Our problem of testing if there is or is not an infinitude of SGP on an interval is in NP (it is not NP-complete) and not in P as it takes an exponential number of evaluations (because the interval is exponential length) to verify the SGP conjecture. The only known algorithm to test for SGPs on an interval is to simply evaluate each integer on that interval which is a brute force type algorithm. This testing of the SGP conjecture gives empirical evidence of if there are an infinite numbers of SGPs or a finite number of SGPs.

2 A Quantum Transformation for the Sophie Germain Prime Conjecture

Since there are efficient classical algorithms to test for primes, such as the AKS primality test, or the Rabin-Miller test this means we can construct an efficient polynomial time and polynomial circuit to implement the following primality indicator function using reversibility e.g. see [1]. Define the following prime indicator function as

$$\begin{aligned} P(x) & \text{ for some } x \geq 0 \\ P(x) &= 1 \text{ if } x \text{ is prime} \\ P(x) &= 0 \text{ if } x \text{ is not prime} \end{aligned} \tag{2}$$

Thus the following function $S'(x)$ can efficiently test if x is a SGP

$$\begin{aligned} S'(x) &= P(x)(2(P(x)) + 1) \text{ for some } x \geq 0 \\ P(x) &= 1 \text{ if } x \text{ is SGP prime} \\ P(x) &= 0 \text{ if } x \text{ is not a SGP prime} \end{aligned} \tag{4}$$

Now if we define the function $S(x)$

$$S(x) = u(S'(x) \text{ OR } S'(x+1) \text{ OR } \dots \text{ OR } S'(\lceil 8 \ln(n) \rceil^2)) \tag{6}$$

Where $u(x)$ is the indicator function

$$\begin{aligned} u(x) &= 1 \text{ if } x > 0 \\ &= 0 \text{ if } x = 0 \end{aligned}$$

note that equation 4 can be computed in polynomial time as it requires $(\lceil 8 \ln(n) \rceil^2 - 1)$ bitwise OR operations on $\lceil 8 \ln(n) \rceil^2$ integers. Now consider the following which evaluates 4 using two quantum registers

$$|x, o\rangle \rightarrow |x, \sum_x S(x+c)\rangle \tag{5}$$

on some interval $[c, c + 2^\alpha - 1]$ for some $\alpha \geq 2$ for the values of x (so $c \leq x \leq c + 2^\alpha - 1$). Then if we assume there are not an infinite number of SGP after some integer d where $d < c$ then all the ket values in the second register are $|0\rangle$. If we assume there is an infinite number of SGP after some integer d where $d < c$ then all the ket values in the

second register are $|1\rangle$. Note we have not assumed that the SGPs do not change density to some intermediate density between $li_2(x)$ and 0 density; this assumption is reasonable as there is no numerical evidence to support otherwise for large values; if this change in density assumption is false then our algorithm does not work properly. Thus we need to transform the above two types of assumed results into definite outputs that corresponds with certainty to if there are an infinite number of SGPs or a finite number of SGPs; the Deutsch-Jozsa does this but it only works for balanced or constant functions. But if instead of 5 we use the version of 5 below

$$|x, 0\rangle \rightarrow |x, \sum_x (xS(x+c)) \bmod 2\rangle \quad (7)$$

which equals

$$|x, 0\rangle \rightarrow |x, \sum_x [(x+c) * u(S'(x+c) \text{ OR } S'(x+1+c) \text{ OR } \dots \text{ OR } S'(c + \lceil 8 \ln(n) \rceil^2))] \bmod 2\rangle \quad (8)$$

...OR $S'(c + \lceil 8 \ln(n) \rceil^2)) \bmod 2\rangle$

Then,now, if we assume there are not an infinite number of SGP after some integer d where $d < c$ then all the values of the kets in the second register still remain as $|0\rangle$ so the output is constant. If we assume is an infinite number of SGP after some integer d where $d < c$ then all the values of ket states in the second register (with some equal amplitude) are $|0\rangle, |1\rangle, |0\rangle, |1\rangle, \dots, |0\rangle, |1\rangle$.-so the output is balanced and periodic. Hence we can use the Deutsch-Jozsa algorithm to find with certainty that on some interval that supports that there is empirical evidence of if there is an infinite number of SGPs on that interval, or there supporting numerical evidence that there are no SGPs of some interval (empirical evidence that the number of SGPs are finite). The Deutsch-Jozsa algorithm is an algorithm to determine if a function is balanced or constant. Classically to determine if a function is balanced or constant on some interval $[c, c+2^\alpha-1]$ takes an (exponential) $2^{\alpha-1}+1$ number of evaluations. In the worst case the best classical algorithm to test for SGPs on the interval $[c, c+2^\alpha-1]$,takes at worst $O(2^{\alpha-1})$ evaluations. This is basically because there are approximately $O(2^{\alpha-1})$ odd integers to evaluate,and the last evaluated odd integer on above interval is enough to show that still maybe infinite SGPs even if all the other odd integers on the interval are not SGPs-though if the SGPs did suddenly take on such a low density then that would be unexpected under current knowledge.

The above works because equation 6 boosts the number of solutions, with 'fake solutions', to simulate a density of 1 for the (logarithmically sized) gaps if the expected density of SGP is as expected; the 'fake solution' only exist if the real solutions exist. If there are no SGPs then there are no 'fake solutions' because there are no real solutions. Note the mod 2 part of equation 7 can be implemented using a single reversible Toffoli acting as an *XOR* gate. The quantum algorithm for SGPs can be modified to verify other conjectures (maybe not related to SGPs), e.g. other prime number type conjectured with certain densities, though it would be need a change (in general) of the $\lceil 8 \ln(n) \rceil^2$ term in 7 and we could estimate the gap size here using Cramer's model. This may result in an algorithm with big-O complexity as exponential or polynomial, or big-O complexity between exponential and polynomial, depending on what exactly the term $\lceil 8 \ln(n) \rceil^2$ need to be replaced with for the other conjecture. Also here we can use Cramer's model to get the version of equations 2 and 3 for the other different conjecture. So we now give the (sketched out) quantum algorithm below.

3 A Quantum Algorithm to Verify if there are an Infinite Number of Sophie Germain Primes

Step 1. Decide the starting point c for the interval $[c, c + 2^\alpha - 1]$ to test the conjecture for the density of SGPs .

Step 2. We have a quantum computer with the register $|x\rangle|y\rangle$. Apply the tensored Hadamards $H^{\otimes n}$ to create the superposition

$$|\Psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^\alpha-1} |x\rangle \quad (12)$$

apply a Hadamard to ancilla qubit.

Step 3. Using the oracle U_f (the oracle with f here is as in the Deutsch-Jozsa algorithm)

$$U|x\rangle|y\rangle = |x\rangle|y \oplus F(x)\rangle \quad (13)$$

where

$$f(X) = x * S(x + c) \mod 2 \quad (14)$$

$$= (x + c) * u(S'(x + c) \text{OR } S'(x + 1 + c) \text{OR } \dots \text{OR } S'(c + \lceil 8 \ln(n) \rceil^2)) \mod 2 \quad (15)$$

$$|x, o \rangle \rightarrow |x, \sum_x (x * S(x + c)) \bmod 2 \rangle \quad (9)$$

which equals

$$|x, o \rangle \rightarrow |x, \sum_x [((x + c) * u(S'(x + c) \text{OR } S'(x + 1 + c) \text{OR } \dots \text{OR } S'(c + \lceil 8 \ln(n) \rceil^2)))] \bmod 2 \rangle \quad (11)$$

the number of states in superposition in the second register will be a power of 2.

Step 4. Apply set of Hadamard gates again to the to the first register. Then measure. If the measurement gives $|0 \rangle^{\otimes n}$ (which is a single measurement on an n bit register) then f is constant which is supporting empirical evidence there are infinite SGPs. Otherwise (any other measurement) different to $|0 \rangle^{\otimes n}$ for the register means that f is a balanced function; hence is supporting evidence that there are finite number of SGPs.

End of algorithm

4 Proof

We prove why the algorithm in the above section works.

The Hadamard gate at input gives the equiprobable type state (at $|\Phi_1 \rangle$ in circuit diagram, Fig. 1, below)

$$= \frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x \rangle (|0 \rangle - |1 \rangle) \quad (20)$$

with x running over the n -bit strings from 0 to $2^n - 1$. Applying the oracle $U_f = U_{\beta(x)}$ gives where $\beta(x) = xS(x + c) \bmod 2$ (where $S(x)$ was defined above as equation 6) gives

$$\frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} |x \rangle (|0 \oplus \beta(0) \rangle - |0 \oplus \beta(1) \rangle) \quad (21)$$

$$= \frac{1}{\sqrt{2^{n+1}}} \sum_{x=0}^{2^n-1} (-1)^{\beta(x)} |x \rangle (|0 \rangle - |1 \rangle) \quad (22)$$

ignoring the ancilla qubit we have

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{\beta(x)} |x\rangle \quad (23)$$

applying a set of n Hadamards to the n qubits acts as a specialised version of the quantum Fourier Transform (where the "angle values" in the unitary matrix of the QFT reduce to ± 1 at some multiple of π) over the Boolean bit strings $\{0, 1\}^n$ and the basis change involves the inner product $x.y$, where the terms of this inner product are computed using *XOR*, the transform is

$$|x\rangle \rightarrow \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x.y} |y\rangle \quad (24)$$

when we apply 24 to 23 we have (at $|\Phi_2\rangle$ or $|\Phi_3\rangle$ in Fig. 1)

$$\sum_{x=0}^{2^n-1} (-1)^{\beta(x)} \left(\frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x.y} |y\rangle \right) \quad (25)$$

Therefore the probability to be in a state k is

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{\beta(x)} (-1)^{x.k} \quad (26)$$

Hence the probability of measuring a string of n zeroes in the first register is

$$\left[\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{\beta(x)} \right]^2 \quad (27)$$

$$= \left[\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{\beta(x)=xS(x+c) \bmod 2} \right]^2 \quad (28)$$

$$= \left[\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{\beta(x)=(x+c)*u(S'(x+c)OR S'(x+1+c)OR...OR S'(c+\lceil 8\ln(n) \rceil^2)) \bmod 2} \right]^2 \quad (29)$$

When $\beta(x)$ is a constant function then 27 or 29 evaluates, when measured, to deterministically a bit string of n zeroes using constructive interference. When $\beta(x)$ is balanced 27 or 29 then the measurement will give a state different to a bit string of n zeroes using destructive interference. The circuit diagram (Fig. 1) is as follows.

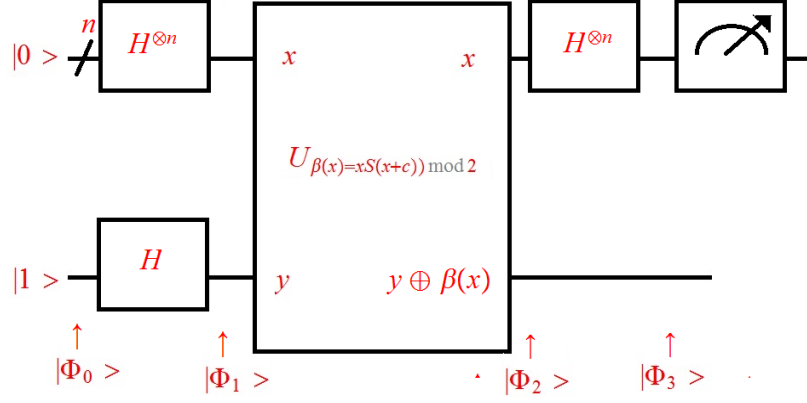


Fig. 1

In a future paper we will give a circuit for our quantum algorithm e.g. using the circuits in [1] for the prime testing circuit or construction $P(x)$ used in equation 4 above.

4.1 Conclusion

We presented a first quantum algorithm that gives an exponential speedup for checking if there are an infinite number of SGPs on the interval $[c, c + 2^\alpha - 1]$; the best classical algorithm takes at worst about $O(2^{\alpha-1})$ evaluations. Remarkably our quantum algorithm takes only $O(1)$ time or one quantum function evaluation which gives an exponential $O(2^{\alpha-1})$ speed up.

References

- [1] J. Latore, Quantum Computation of Prime Number Functions, 2013, ArXiv Preprint